

BUTT OF LEWIS LH, United Kingdom

WMO: 030250

Lat: 58.517N

Lon: 6.267W

Elev: 23

StdP: 101.05

Time Zone: 0.00 (EUW)

Period: 86-97

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.2	1.0	-4.5	2.6	2.8	-3.4	2.8	3.2	25.0	6.5	23.2	6.2	5.7	210	0.952

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.3	17.3	14.8	16.1	14.2	15.2	13.6	15.3	16.7	14.5	15.7	13.8	14.9	6.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	10.4	15.8	13.9	9.9	15.1	13.3	9.5	14.5	42.9	16.9	40.8	15.8	39.0	14.9	18.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.5	19.2	17.6	DB	-1.7	21.8	0.9	1.2	-2.3	22.7	-2.9	23.4	-3.4	24.0	-4.0	24.9	(4)
(5)			WB	-2.5	17.3	1.0	0.5	-3.2	17.6	-3.8	17.9	-4.4	18.2	-5.1	18.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	5.3	4.9	5.4	6.6	9.1	10.9	12.8	12.6	11.2	9.3	7.4	5.9		
	DBStd	3.35	1.99	1.98	2.01	1.78	2.10	1.87	1.56	1.38	1.63	1.94	1.90	2.09		
	HDD10.0	836	145	142	142	103	43	11	0	6	37	79	128			
	HDD18.3	3600	403	376	400	353	287	223	172	178	213	282	328	386		
	CDD10.0	279	0	0	0	1	14	37	87	81	43	14	2	1		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	8.5	10.6	10.0	10.3	8.0	7.6	6.4	6.7	6.9	8.2	8.8	9.3	9.8		
(15) Precipitation	PrecAvg	1331	155	120	115	81	73	72	84	93	114	137	139	148		
	PrecMax	1722	285	278	234	123	161	126	157	204	190	222	233	276		
	PrecMin	959	59	55	39	33	10	33	30	41	37	50	68	32		
	PrecStd	169	55	63	45	21	35	27	29	38	39	48	36	60		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	10.2	11.0	13.0	18.3	18.4	19.4	18.6	16.8	14.1	12.2	11.6		
		MCWB	9.9	9.1	9.9	10.1	14.5	14.6	16.2	15.7	14.7	12.3	11.5	10.4		
	2%	DB	10.0	9.5	9.8	11.0	15.5	16.1	17.1	16.4	15.1	13.2	11.4	10.5		
		MCWB	9.2	8.8	8.9	9.2	12.7	13.7	15.0	14.7	13.5	11.7	10.6	9.9		
	5%	DB	9.3	8.7	9.1	10.1	13.3	14.5	16.0	15.3	14.2	12.5	10.7	9.9		
		MCWB	8.5	8.1	8.2	8.7	11.0	12.9	14.3	13.8	12.8	11.2	9.7	9.4		
	10%	DB	8.6	7.9	8.3	9.2	11.8	13.4	15.2	14.6	13.6	12.0	10.2	9.2		
		MCWB	7.8	7.1	7.4	8.0	10.0	11.9	13.7	13.2	12.3	10.8	9.1	8.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	9.6	10.1	11.0	14.7	15.3	16.6	16.2	15.5	13.0	11.9	10.7		
		MCDB	10.5	9.8	10.5	12.5	17.8	17.3	18.5	17.7	16.4	13.6	12.1	11.2		
	2%	WB	9.4	9.0	9.1	9.7	12.9	14.0	15.3	15.1	14.1	12.1	10.9	10.1		
		MCDB	9.8	9.3	9.7	10.5	15.1	15.9	16.7	16.1	14.7	12.7	11.2	10.4		
	5%	WB	8.7	8.2	8.4	8.9	11.3	13.1	14.6	14.2	13.3	11.6	10.2	9.5		
		MCDB	9.2	8.7	9.0	9.8	12.9	14.4	15.7	15.0	13.9	12.3	10.7	9.9		
	10%	WB	7.9	7.3	7.6	8.1	10.2	12.2	13.9	13.6	12.6	11.0	9.4	8.6		
		MCDB	8.5	7.9	8.2	9.0	11.3	13.2	14.9	14.4	13.4	11.9	10.1	9.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.5	3.4	3.9	3.6	3.5	3.2	3.3	3.4	3.6	3.4	3.4	3.5		
		MCDBR	4.0	3.6	4.4	4.7	6.5	5.2	4.9	4.5	4.5	3.7	3.5	3.6		
	5% WB	MCWBR	4.3	3.9	4.3	3.7	4.4	3.8	3.4	3.5	3.8	3.4	3.8	4.0		
		MCWBR	3.9	3.4	4.2	4.0	5.5	5.0	4.7	4.4	4.1	3.4	3.5	3.7		
(40) Clear-Sky Solar Irradiance	taub	0.352	0.356	0.369	0.383	0.380	0.372	0.377	0.368	0.365	0.350	0.327	0.317			
	taud	2.100	2.235	2.301	2.321	2.386	2.451	2.452	2.483	2.454	2.355	2.197	2.091			
	Ebn at Noon	426	650	766	822	853	865	851	834	772	657	464	343			
	Edh at Noon	55	76	94	107	108	103	100	91	81	67	48	40			
(44) All-Sky Solar Radiation	RadAvg	0.31	0.89	2.00	3.54	4.73	4.77	4.42	3.58	2.34	1.18	0.44	0.21			
	RadStd	0.04	0.09	0.13	0.23	0.34	0.41	0.30	0.23	0.12	0.10	0.04	0.02			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page